MIL-A-47282A(MI)
30 May 1975
SUPERSEDING
MIL-A-47282(MI)
9 August 1974
MIS-13197A
20 February 1975

MILITARY SPECIFICATION

ADHESIVE, PASTE, HEAT RESISTANT

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

.1. Scope. This specification covers the requirements for a heat resistant paste adhesive for use in bonding various materials. The bond line will be subject in use to short time exposures at temperatures of 600 degrees Fahrenheit (F) and above.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/5	Aluminum Alloy Alclad 2024, Plate and Sheet
PPP-C-96	Can, Metal, 26 Gage and Lighter

Military

*	MIL-A-9067	Adhesive Bonding, Process and Requirements for
*	MIL-A-47281(MI)	Adhesive Primer, Heat Resistant

FSC 8040

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:07Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-175	Adhesive, Methods of Testing

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM D 1002	Strength Properties of Adhesive in Shear by Tension Loading (Metal to Metal)

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample or samples, as required, shall be tested and shall meet the requirements of this specification.

3.2 Materials. The adhesive shall be a two-component epoxy-phenolic material with one component being the resin base material designated as Component A and the other component being the curing agent designated as Component B.

3.2.1 Fillers. Fillers may be incorporated in either Component A or Component B. Any fillers shall be finely divided and remain uniformly dispersed and suspended throughout the adhesive during its normal pot life. The fillers shall be readily dispersible throughout either component, where supplied as a part of the component, during the entire usable storage life of the component.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:07Z
Check the source to verify that this is the current version before use.

NOTE: The fillers shall be highly corrosion and moisture resistant and shall withstand the maximum temperatures to which the adhesive is subjected both during its cure and in end use.

* 3.2.2 Solids content. The solids content of the two components when mixed shall be a minimum of 92 percent.

3.2.3 Form. The two components, A and B, shall mix readily to a smooth paste free of lumps. The paste shall have a consistency suitable for application by trowel, spatula, or stiff brush and shall be thixotropic in nature. The components shall not settle out or separate during the course of the adhesives' normal pot life.

3.3 Working characteristics.

3.3.1 Application. The adhesive shall be capable of being readily applied in accordance with the manufacturer's instructions (see 3.5) at room temperatures at least between plus 60 and 100 degrees F and at relative humidities of at least up to 75 percent unless the manufacturer's instructions specifically require heating of either components or mixed adhesive, for mixing or application.

3.3.2 Pot life. The minimum pot life of the adhesive, ready and use for application, shall be 8 hours at 77 plus 15 minus 5 degrees F.

3.3.3 Curing. The adhesive shall not require more than 5 hours for curing at a temperature not to exceed 360 degrees F. The pressure required for proper curing shall not exceed 50 pounds per square inch (psi). The following cure schedule shall be used:

160 degrees F for 1 hour, plus
275 degrees F for 1 hour, plus
350 degrees F for 2 hours

Cool slowly to 200 degrees F maximum before relieving pressure.

NOTE: Temperature tolerance: Plus or minus 10 degrees F
Time Tolerance: Plus or minus 5 minutes

3.3.4 Storage life. The adhesive shall have a storage or shelf life of not less than 6 months from the date of shipment when Part A is stored in the original unopened containers at a temperature of 40 degrees F or below and Part B is stored in the original unopened containers at a
* temperature of 85 degrees F or below. Useful shelf life is indicated by the ability of the adhesive to meet the 500 plus or minus 10 degrees F strength requirements of Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:07Z
Check the source to verify that this is the current version before use.

3.4 Mechanical properties. The adhesive system shall meet the lap shear strength requirements of Table I when cured per 3.3.3. Values shall represent an average of five tests. Elevated temperatures shall be obtained using a circulating air oven.

Table I. Mechanical properties

*Test Temperature, F ± 10	75	500
Lap Shear Strength, psi, minimum	1500	1300

*Soak time for 500 degrees F shall be 10 plus or minus 1 minute. Oven bring up time for the test specimen shall not exceed 5 minutes.

3.5 Instruction sheet. The manufacturer shall provide a coded and titled instruction sheet with each container in each shipment of adhesive outlining instructions for its use to the extent that the following minimum information shall be given:

a. Proportions of Component A to Component B even if supplied in "kit" form.

b. Any special mixing instructions that must be followed, such as temperature or humidity restrictions, special mixing equipment required, or special indicators to be observed for complete mixing. If no special mixing instructions are required, mixing instructions shall be limited to (a), above.

c. Safety precautions to be observed in the use of the adhesive or components, if applicable.

d. Storage requirements.

NOTE: The above information may be incorporated on the container labels (see 5.2) in lieu of a separate instruction sheet.

3.6 Identification of product. The manufacturer shall identify the product submitted under this specification by a trade name and code number.

3.7 Workmanship. Both components shall be free of foreign matter and shall be prepared in accordance with the best commercial practices for this material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:07Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities * or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods for the preparation of subsequent lots of adhesive. Preproduction samples which do not meet all the requirements of this specification shall be subject to rejection.

4.2.1 Sample size. A two container unit consisting of component A and Component B, in their proper mixing ratios, shall be supplied to provide approximately one quart of mixed adhesive.

4.2.2 Re-examination and retest. Re-examination of a product which has previously passed the preproduction tests shall be carried out with a preproduction sample, per this specification, under any of the following conditions:

a. The manufacturer has modified his product or changed his material or processing sufficiently so that the validity of previous preproduction tests is questionable.

b. The requirements in the specification have been amended or revised sufficiently to affect the character of the product.

c. When deemed necessary to determine that the quality of the product is being maintained.

4.3 Classification of inspection. The examination and testing of the adhesive shall be classified as follows:

a. Preproduction inspection. 4.3.1

b. Quality conformance inspection. 4.3.2

4.3.1 Preproduction inspection. Preproduction inspection shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:07Z
Check the source to verify that this is the current version before use.

4.3.2 Quality conformance inspection. Quality conformance inspection for acceptance of the adhesive shall consist of the following examinations and tests:

a.	Solid content	4.4.2
b.	Mechanical properties at 500 degrees F	4.4.4

4.3.2.1 Lot size. Lot size shall consist of all adhesive manufactured at one time, from one batch, forming part of one contract or order, and submitted for acceptance at one time and place. A batch is defined as that quantity of material processed through a mill or mixer at one time.

4.3.2.2 Sampling. Unless otherwise specified (see 6.2), the sample shall be selected at random from the inspection lot and shall be a two container unit consisting of Component A and Component B in their proper mixing ratio. Failure of the sample to meet the quality conformance requirements shall be cause for lot rejection.

4.4 Test methods.

4.4.1 Examination of product. Components A and B singly or when mixed shall be visually examined for conformance with the requirements of 3.2, 3.2.1, 3.2.3 and 3.3.1.

4.4.2 Solids content. The solids content shall be determined in accordance with Method 4021, Procedure B of FED-STD-175 and shall meet the requirements of 3.2.2.

NOTE: An alternate oven cure cycle of 1 hour at 335 plus or minus 5 degrees F may be used.

4.4.3 Pot life. Pot life (3.3.2) shall be determined as the time elapsed between mixing Components A and B and that time after which the material can no longer be readily applied per 3.3.1.

4.4.4 Mechanical properties. The adhesive system shall meet the * minimum strenths listed in Table I, when prepared and tested in accordance with ASTM D 1002 with the following exceptions and/or additions:

a. Adherends: 2024-T3 (QQ-A-250/5) aluminum, 0.063 inch thick.

* b. Surface cleaning: Vapor grease, grit blast with 220 grit aluminum oxide using clean, dry, oil-free air or gaseous nitrogen and tap to remove adhering grit or blow clean with clean, dry, oil-free air or gaseous nitrogen. Alternately, the cleaning of the adherends after degreasing may be done using the sulfuric acid-sodium dichromate etch procedure given in MIL-A-9067.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:07Z
Check the source to verify that this is the current version before use.

c. Priming: Within 2 hours after cleaning, prime the surface with MIL-A-47281(MI) primer, applied in accordance with the manufacturer's instructions. Allow the primer to air dry a
* minimum of 30 minutes followed by an oven dry in a forced air circulating oven for 60 plus or minus 10 minutes at 150 plus or minus 10 degrees F.

d. Within 24 hours after priming, apply the adhesive mixed per the manufacturer's instructions to each primed surface and assemble the details. A glass thread in the bond line or
* shims may be used to insure a bond line thickness as recommended by the manufacturer and to prevent excessive squeeze-out of the adhesive during cure.

e. Cure: Using 25-50 psi contact pressure, cure the specimens per the cure schedule of 3.3.3 and 3.4.

4.5 Storage. After 6 months have expired, based on the date of shipment and storage temperatures of 40 degrees F or below for Part A and 85 degrees F or below for Part B, a random sample of the adhesive
* shall be selected (4.3.2.2) and when tested per this specification it shall meet the 500 degrees F requirement of Table I. Thereafter, the adhesive shall be retested at every 4 month interval or prior to use.

4.6 Workmanship. Components A and B shall be visually examined for evidence of dirt, undissolved matter and other defects.

4.7 Preservation, packaging, packing, and marking. The preservation, packaging, packing and marking shall be examined to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be as specified herein.

5.1.1 Packaging. Components A and B shall be packaged in metal containers conforming to PPP-C-96. The type, class, shape, and size of the container and type of closure, when required, shall be as specified by the procuring activity (see 6.2).

5.1.2 Packing. The adhesive shall be packaged in two container units or "kits". The amount of one component packaged in one container (5.1.1 of a kit shall be the correct amount to be mixed with the amount of the component packaged in the other container (5.1.1) of that kit. The material packaged as specified above shall be packed in exterior type containers, in a manner that will insure safe transporation at the lowest rate to the point of delivery. Containers shall meet common carrier regulations, as applicable to the mode of transportation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:07Z
Check the source to verify that this is the current version before use.

5.1.3 Marking. Interior and exterior packages shall be marked in accordance with MIL-STD-129. Exterior containers shall also be marked with a note relative to storage such as:

a. Materials conforming to Part A of this specification should be stored at a temperature of 40 degrees F or less in the absence of direct sunlight.

b. Materials conforming to Part B of this specification should be stored at a temperature of 85 degrees F or less in the absence of direct sunlight.

c. If transit or normal storage conditions necessitate exposure of the materials to temperature greater than specified in (a) and (b) the storage life of the respective material will be proportionally reduced.

5.2 Containers. Each container of component A or B, shall be permanently labeled with at least the following information:

a. Adhesive, elevated temperature resistant, Component A or B (as applicable).

b. This specification number.

c. Manufacturer's identification of adhesive.

d. Batch number.

e. Date of shipment.

f. Manufacturer's name.

g. Procuring activity P.O. Number.

h. Mixing and usage directions (see 3.5).

i. Lot number.

6. NOTES

6.1 Intended use. The material conforming to this specification is intended for use as an elevated temperature resistant adhesive in ablative or metal constructions in small rocket engines.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:07Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a proproduction sample is required, and if so, pertinent details (see 3.1).

c. Sampling plan, if other than specified (see 4.3.2.2).

d. Requirements for preservation, packaging, packing, and marking, if different (see 5.1).

e. Applicable level(s) of preservation, packaging, and packing.

f. Quantity, size and type of containers required (5.1.1).

6.3 Certification of conformance. The supplier shall present a certificate of conformance with each inspection lot stating that the adhesive meets the requirements of 3.2.2 and Table I. In addition, the supplier shall certify that the formulation of the adhesive as designated by a certain trade name and/or proprietary number has not been changed since the preproduction sample was approved under this specification.

6.4 Supersession data. This specification includes the requirements of military specification MIL-A-47282(MI) dated 9 August 1974 and Missile Interim Specification MIS-13197A dated 20 February 1975.

* 6.5 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - MI

Preparing Activity
Army - MI

Project No. 8040-A075

☆ U. S. GOVERNMENT PRINTING OFFICE: 1975-603-119/4933.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:07Z
Check the source to verify that this is the current version before use.